

Concentration for Random Product Formulas

Course Paper — Theory and Numerical Experiments

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Abstract

This part reconstructs the theory of *Concentration for random product formulas* (Chen, Huang, Kueng, Tropp), which analyzes the qDRIFT randomized product formula for Hamiltonian simulation. The paper asks whether qDRIFT's gate count — famously free of explicit dependence on the number L of Hamiltonian terms — comes from *averaging* many random circuits or is already present in *a single* one. The answer is the latter: one realization typically approximates the ideal evolution with high probability, using $\tilde{O}(n\lambda^2 t^2/\epsilon^2)$ gates for a worst-case (diamond-norm) guarantee and $\tilde{O}(\lambda^2 t^2/\epsilon^2)$ for a fixed input, where n is the qubit number and $\lambda = \sum_j \|h_j\|$. The proof decomposes the error into a deterministic bias and a random fluctuation, controlling the latter by a matrix martingale (worst case) or a dimension-free vector martingale (fixed input).

1 Setup and the Central Question

For $H = \sum_{j=1}^L h_j$ on n qubits, digital simulation approximates $U = e^{-itH}$ by a product $V_N \cdots V_1$ of simple gates, with cost measured by N . The first-order Lie–Trotter formula $e^{-itH} \approx (e^{-i(t/r)h_L} \cdots e^{-i(t/r)h_1})^r$ uses $N = rL$ gates: each layer must cycle through all L terms, an explicit L -dependence that is costly when L is large.

qDRIFT (Campbell) avoids this. With $\lambda = \sum_j \|h_j\|$ and $p_j = \|h_j\|/\lambda$, each step samples one index $j \sim p_j$ and applies

$$V_k = \exp\left(-i \frac{t}{N} X_k\right), \quad X_k = \frac{\lambda}{\|h_j\|} h_j \text{ w.p. } p_j, \quad \text{so} \quad \mathbb{E}[X_k] = \sum_j p_j \frac{\lambda}{\|h_j\|} h_j = H. \quad (1)$$

This unbiasedness gives $\mathbb{E}[V_k] \approx e^{-i(t/N)H} = U^{1/N}$, hence by independence $\mathbb{E}[V_N \cdots V_1] = (\mathbb{E}V)^N \approx U$. Campbell proved the *averaged channel* $\bar{\mathcal{V}}(\rho) = \mathbb{E}[V_N \cdots V_1 \rho V_1^\dagger \cdots V_N^\dagger]$ is ϵ -close to $\mathcal{U}(\rho) = U \rho U^\dagger$ in diamond distance once $N = O(\lambda^2 t^2/\epsilon)$, with *no* explicit L .

That bound concerns the average over circuits, but a user runs the box once and gets a single unitary $V_N \cdots V_1$. The paper asks: *does one random circuit already work?* This separates *random sampling* from *averaging*. The answer: sampling alone removes the L -dependence; averaging only improves the accuracy exponent from $1/\epsilon^2$ to $1/\epsilon$.

Throughout, $d = 2^n$, the generator $(t/N)X_k$ is Hermitian with $\|(t/N)X_k\| = t\lambda/N$ (since $\|X_k\| = \lambda$), and $\mathcal{V}^{(N)}(\rho) = V_N \cdots V_1 \rho V_1^\dagger \cdots V_N^\dagger$ is the single-realization channel. Two error notions arise. For *all* inputs and observables (worst case), the relevant quantity for unitary channels is the diamond distance,

$$\frac{1}{2} \|\mathcal{U} - \mathcal{V}\|_\diamond = \max_{|\psi\rangle} \frac{1}{2} \|U|\psi\rangle\langle\psi|U^\dagger - V|\psi\rangle\langle\psi|V^\dagger\|_1 = \max_{|\psi\rangle} \max_{\|O\| \leq 1} |\langle O \rangle_{U|\psi\rangle} - \langle O \rangle_{V|\psi\rangle}|; \quad (2)$$

for a single *fixed* input ρ , it is the trace distance $\frac{1}{2}\|U\rho U^\dagger - V\rho V^\dagger\|_1$.

2 Main Results

Theorem 1 (Worst case, diamond norm). *Draw $V^{(N)} = V_N \cdots V_1$ from (1) with $N \geq \Omega((n + \log(1/\delta))\lambda^2 t^2 / \epsilon^2)$. Then with probability $\geq 1 - \delta$, $\max_{|\psi\rangle} \max_{\|O\| \leq 1} |\langle O \rangle_{V^{(N)}|\psi\rangle} - \langle O \rangle_{U|\psi\rangle}| < \epsilon$. Moreover $\mathbb{E}[\frac{1}{2}\|\mathcal{U} - \mathcal{V}^{(N)}\|_\diamond] \lesssim \sqrt{n\lambda^2 t^2 / N}$.*

Theorem 2 (Fixed input, trace norm). *Fix any state $|\psi\rangle$ and draw $V^{(N)}$ with $N \geq \Omega(\lambda^2 t^2 \log(1/\delta) / \epsilon^2)$. Then with probability $\geq 1 - \delta$, $V^{(N)}|\psi\rangle$ is ϵ -close to $U|\psi\rangle$ in trace distance, and $\max_\rho \mathbb{E}[\frac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1] \lesssim \sqrt{\lambda^2 t^2 / N}$.*

Neither bound contains L , so QDRIFT’s L -independence already holds for one circuit; by the probabilistic method Theorem 1 even proves such L -free formulas *exist*. The price for a single circuit is accuracy $1/\epsilon^2$ (vs. $1/\epsilon$ for the average), and the worst case carries a factor $n = \log_2 d$. Strikingly, fixing the input removes n *without any knowledge of the state* — only the assumption that $|\psi\rangle$ is chosen independently of the gates; deterministic methods usually need spectral structure (e.g. a low-energy subspace) for such a gain. Writing $\tilde{O}$ for log-suppressed factors, the hierarchy is

$$N_{\text{avg}} = O\left(\frac{\lambda^2 t^2}{\epsilon}\right) \prec N_{\text{fixed}} = \tilde{O}\left(\frac{\lambda^2 t^2}{\epsilon^2}\right) \prec N_{\text{worst}} = \tilde{O}\left(\frac{n\lambda^2 t^2}{\epsilon^2}\right).$$

The $1/\epsilon$ vs. $1/\epsilon^2$ gap is the “quadratic improvement from mixing,” an instance of Jensen’s inequality for the convex distance functional.

3 Proof Strategy

All regimes share one skeleton: reduce the diamond distance to an operator (or vector) norm, split the error as

$$\|V_N \cdots V_1 - U\| \leq \underbrace{\|(\mathbb{E}V)^N - U\|}_{\text{deterministic bias}} + \underbrace{\|V_N \cdots V_1 - (\mathbb{E}V)^N\|}_{\text{random fluctuation}}, \quad (3)$$

bound the bias by matrix-exponential estimates plus telescoping, and bound the fluctuation by a martingale inequality. Geometrically $V_N \cdots V_1$ is a random walk on $U(d)$: the bias is the offset of its mean from U , the fluctuation its spread about the mean.

Step 1: diamond $\rightarrow$ operator norm. For unitary channels no ancilla is needed, so the diamond distance collapses to a norm.

Lemma 1. *For unitary $\mathcal{U}, \mathcal{V}$ and any pure $|\psi\rangle$, $\frac{1}{2}\|\mathcal{U}(|\psi\rangle\langle\psi|) - \mathcal{V}(|\psi\rangle\langle\psi|)\|_1 \leq \|(U - V)|\psi\rangle\|_{\ell_2}$, hence $\frac{1}{2}\|\mathcal{U} - \mathcal{V}\|_\diamond \leq \|U - V\|$; for the averaged channel, $\frac{1}{2}\|\mathcal{U} - \mathbb{E}[\mathcal{V}]\|_\diamond \leq 2\|U - \mathbb{E}V\|$.*

Proof idea. Fix $|\psi\rangle$ and set $|u\rangle = U|\psi\rangle$, $|v\rangle = V|\psi\rangle$, so $|\langle u, v \rangle| \leq 1$. The Fuchs–van de Graaf relation converts the output trace distance into a fidelity:

$$\frac{1}{2}\| |u\rangle\langle u| - |v\rangle\langle v| \|_1 = \sqrt{1 - |\langle u, v \rangle|^2} = \sqrt{(1 + |\langle u, v \rangle|)(1 - |\langle u, v \rangle|)} \leq \sqrt{2(1 - \text{Re}\langle u, v \rangle)} = \| |u\rangle - |v\rangle \|_{\ell_2}.$$

Maximizing over pure inputs, and using that stabilization (an ancilla) is unnecessary for the diamond distance of unitary channels, gives the first two claims. For random V with $E = \mathbb{E}V$, expand $\mathbb{E}[V|\psi\rangle\langle\psi|V^\dagger] = E|\psi\rangle\langle\psi|E^\dagger + \mathbb{E}[(V - E)|\psi\rangle\langle\psi|(V - E)^\dagger]$ (cross terms vanish); bounding the bias term and the positive-semidefinite variance term separately, each by $\|(U - E)|\psi\rangle\|_{\ell_2}$, yields the factor 2 (due to Allen–Zhao, correcting an earlier claim that omitted it). $\square$

Step 2: deterministic bias. Using $\|e^{iX} - I\| \leq \|X\|$ and $\|e^{iX} - iX - I\| \leq \frac{1}{2}\|X\|^2$ for Hermitian X , with $\|(t/N)X_k\| = t\lambda/N$ and $\mathbb{E}[(t/N)X_k] = (t/N)H$:

$$\|V - \mathbb{E}V\| \leq \frac{2t\lambda}{N}, \quad \|\mathbb{E}V - U^{1/N}\| \leq \left(\frac{t\lambda}{N}\right)^2. \quad (4)$$

The first follows from $\|V - \mathbb{E}V\| \leq \|e^{-iX} - I\| + \mathbb{E}\|I - e^{-iX}\| \leq 2\|X\|$ (with $X = (t/N)X_k$); the second from $\|\mathbb{E}V - U^{1/N}\| \leq \frac{1}{2}\mathbb{E}\|X\|^2 + \frac{1}{2}\|\mathbb{E}X\|^2 \leq \mathbb{E}\|X\|^2$ via the first-order estimate and Jensen. The single-step bias accumulates by a telescoping bound $\|(\mathbb{E}V)^N - (U^{1/N})^N\| \leq N\|\mathbb{E}V - U^{1/N}\|$ (iterate $\|A_1A_2 - B_1B_2\| \leq \|A_1 - B_1\| + \|A_2 - B_2\|$ for norm- ≤ 1 factors), giving the key estimate

$$\|U - \mathbb{E}[V_N \cdots V_1]\| = \|(U^{1/N})^N - (\mathbb{E}V)^N\| \leq \frac{t^2\lambda^2}{N}. \quad (5)$$

This is exactly the averaged-channel bias, $\leq \epsilon/2$ once $N \geq 2t^2\lambda^2/\epsilon$ — already L -free with the favorable $1/\epsilon$ rate. Geometrically, the bias is how far the random walk’s *mean* $(\mathbb{E}V)^N$ sits from the target U ; the next step controls the walk’s *spread* about that mean, and the averaged channel pays only the bias — the root of its quadratic accuracy advantage.

Step 3: fluctuation via a matrix martingale. Interpolate between the two ends of (3) by

$$B_k = (\mathbb{E}V)^{N-k}V_k \cdots V_1, \quad B_0 = (\mathbb{E}V)^N, \quad B_N = V_N \cdots V_1. \quad (6)$$

Adjacent iterates differ only at slot k , where $\mathbb{E}V$ is replaced by a fresh V_k . The process is causal and obeys $\mathbb{E}[B_k | V_{k-1} \cdots V_1] = (\mathbb{E}V)^{N-k}\mathbb{E}[V_k]V_{k-1} \cdots V_1 = B_{k-1}$, so it is a martingale. Its increments satisfy $\|C_k\| = \|(\mathbb{E}V)^{N-k}(V_k - \mathbb{E}V)V_{k-1} \cdots V_1\| \leq \|V_k - \mathbb{E}V\| \leq 2t\lambda/N =: R$ and predictable variation $\|\sum_k \mathbb{E}_{k-1} C_k C_k^\dagger\| \leq NR^2 =: v$. Matrix Freedman, $\Pr[\|B_N - B_0\| \geq \tau] \leq 2d \exp(\frac{-\tau^2/2}{v+R\tau/3})$, then gives:

Proposition 1. For $\epsilon \in [0, 4t\lambda]$, $\Pr[\|V_N \cdots V_1 - \mathbb{E}[V_N \cdots V_1]\| \geq \epsilon/2] \leq 2d \exp(-N\epsilon^2/44t^2\lambda^2)$. Hence $N \geq (44t^2\lambda^2/\epsilon^2) \log(2d/\delta)$ forces the deviation $\leq \epsilon/2$ w.p. $\geq 1 - \delta$.

The error is sub-Gaussian with variance $\sim N(t\lambda/N)^2$, a random walk of N steps of size $t\lambda/N$. The prefactor $2d$ is intrinsic to a *matrix*-valued martingale; under the log it becomes $\log(2d) = O(n)$ — the sole source of the system-size factor. Combining (5) ($\leq \epsilon/2$ for $N \geq 2t^2\lambda^2/\epsilon$) with Proposition 1 ($\log(2d/\delta) = O(n + \log \frac{1}{\delta})$) proves Theorem 1; integrating the tail gives the expected bound.

Step 4: fixed input via a vector martingale. For a fixed $|\psi\rangle$ one compares the output vectors $U|\psi\rangle$ and $V_N \cdots V_1|\psi\rangle$, so (6) becomes the vector martingale $|\psi_k\rangle = V_k \cdots V_1|\psi\rangle$. Vector concentration is *dimension-free*; the tool is uniform smoothness of the ℓ_2 norm: if $\mathbb{E}[y | x] = 0$ then $(\mathbb{E}\|x + y\|_{\ell_2}^q)^{2/q} \leq (\mathbb{E}\|x\|_{\ell_2}^q)^{2/q} + (q-1)(\mathbb{E}\|y\|_{\ell_2}^q)^{2/q}$ for $q \geq 2$. Applied recursively across the increments (whose two pieces are conditionally orthogonal) with $\mathbb{E}\|V_k - \mathbb{E}V_k\|^q \leq (2t\lambda/N)^q$, it yields $(\mathbb{E}\|(V_N \cdots V_1 - \mathbb{E}[\cdots])|\psi\rangle\|_{\ell_2}^q)^{1/q} \leq 2\sqrt{(q-1)(t\lambda)^2/N}$. With the negligible bias (5) and Markov’s inequality (optimizing q), for $N \geq (t\lambda)^2$,

$$\max_{\rho} \mathbb{E}[\tfrac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1] \leq 4\sqrt{\tfrac{t^2\lambda^2}{N}}, \quad \max_{\rho} \Pr[\tfrac{1}{2}\|\mathcal{U}(\rho) - \mathcal{V}^{(N)}(\rho)\|_1 > \epsilon] \leq \exp(-\tfrac{\epsilon^2 N}{32e t^2 \lambda^2}), \quad (7)$$

which is Theorem 2. The absence of d here is precisely why the count is n -free: the dimension-free vector inequality removes the n that Theorem 1 pays to cover all 2^n inputs at once.

4 Intuition and Generalization

Why the n appears (and is necessary). For the diagonal commuting Hamiltonian $H = 2^{-n} \sum_{p \in \{0,1\}^n} \alpha_p Z_p$ ($\alpha_p \in \{\pm 1\}$), $e^{-iH}|b\rangle = e^{-iS(b)}|b\rangle$ with phase $S(b) = 2^{-n} \sum_p c_p(b)$, $c_p(b) \in \{\pm 1\}$. Sampling N terms gives the Monte Carlo estimate $\hat{S}(b) = \frac{1}{N} \sum_k c_{p_k}(b)$, which by Hoeffding concentrates around $S(b)$ with deviation $\sim 1/\sqrt{N}$. For *one* basis state, $N = 1/\epsilon^2$ suffices (matching Theorem 2); to control *all* 2^n states, a union bound multiplies the failure probability by 2^n , so $N = n/\epsilon^2$ is needed (matching Theorem 1). Thus $n = \log(2^n)$ is the log-count of states controlled at once, the same role $\log(2d)$ plays in Proposition 1. The dependence is tight at both extremes: single-site $H = \sum_k Z_k$ gives, via a multinomial central-limit argument, a matching lower bound $\mathbb{E}\|U - V_N \cdots V_1\| \gtrsim \sqrt{(n-1)(t\lambda)^2/N}$, while for the all-strings $H = \sum_p Z_p$ the diagonal entries become asymptotically independent (a Walsh–Hadamard/CLT analysis), making the union bound tight and again forcing $\sqrt{n}$. Thus commuting Hamiltonians saturate the upper bound, so the factor n is not a proof artifact.

Beyond qDRIFT. The proof used only two properties of the random factors V_k approximating $U = U_N \cdots U_1$: *causality* (each V_k depends on the past, not the future) and *accurate steps*, $\|V_k - \mathbb{E}_{k-1} V_k\| \leq a_k$, $\|\mathbb{E}_{k-1} V_k - U_k\| \leq b_k$. The same analysis then gives $\|\mathcal{U} - \mathcal{V}\|_{\diamond} \leq 2 \sum_k a_k$ (worst case), $\mathbb{E}\|\mathcal{U} - \mathcal{V}\|_{\diamond} \lesssim \sqrt{n \sum_k a_k^2} + 2 \sum_k b_k$ (typical), $\mathbb{E}\|\mathcal{U}(\rho) - \mathcal{V}(\rho)\|_1 \lesssim \sqrt{\sum_k a_k^2} + 2 \sum_k b_k$ (fixed input), and $\|\mathcal{U} - \mathbb{E}\mathcal{V}\|_{\diamond} \leq 2 \sum_k b_k$ (average). The bias is governed by $\sum_k b_k$, the fluctuation by $\sqrt{\sum_k a_k^2}$, with $\sqrt{n}$ only in the worst-case channel norm. This covers randomly permuted Suzuki formulas and randomized compiling. A by-product is a sharpened mixing lemma, $\|\mathcal{U}_k - \mathbb{E}\mathcal{V}_k\|_{\diamond} \leq 4\|U_k - \mathbb{E}V_k\|$, removing the $\frac{1}{2}a^2$ term of the Hastings–Campbell version.

Two conceptual remarks. First, the $1/\epsilon$ vs. $1/\epsilon^2$ split mirrors the distinction between *weak* and *strong* error for stochastic differential equation solvers: the averaged channel tracks $\mathbb{E}[V_N \cdots V_1]$ (a weak, mean quantity, convergence order 1 in $1/N$), whereas a single realization must control $V_N \cdots V_1$ itself (a strong, pathwise quantity, order $1/2$). Second, the guarantees are *existential*: Theorem 1 shows that a good circuit is sampled with high probability, but

certifying that a given drawn circuit is accurate is a separate, harder problem — a gap that motivates a posteriori estimators and derandomized (low-discrepancy) constructions. Because the count is quadratic in t , qDRIFT is most attractive for short-time simulation and for Hamiltonians with many terms, where deterministic formulas pay the explicit L .

5 Numerical Experiments

This section complements the theoretical discussion with direct numerical experiments. The goal is not to fit the constants in the concentration bounds, but to reproduce their qualitative message: a single qDRIFT product formula has a larger error when it must work for all input states, while the fixed-input error is nearly independent of the system size once the Hamiltonian strength λ is held fixed.

5.1 qDRIFT protocol and error metrics

For each Hamiltonian $H = \sum_j h_j$, the experiments use the qDRIFT sampling rule from Eq. (1). At each step, one term is sampled with probability $p_j = \|h_j\|/\lambda$, where $\lambda = \sum_j \|h_j\|$, and the short-time unitary

$$V_k = \exp\left(-i \frac{t}{N} \frac{h_j}{p_j}\right)$$

is applied. Each plotted point is the average over 50 independent random product formulas. The evolution time is fixed to $t = 2$.

Two errors are measured. For the worst-case setting, the experiment computes the operator norm

$$\epsilon_{\text{all}} = \|U - V_N \cdots V_1\|_{\infty},$$

which controls all input states and all observables through Lemma 1. For the fixed-input setting, the input state is a tensor product of single-qubit Haar-random states, chosen independently of the qDRIFT gates, and the experiment computes

$$\epsilon_{\text{fixed}} = \|U|\psi\rangle - V_N \cdots V_1|\psi\rangle\|_{\ell_2}.$$

The relative-size plots normalize each curve by its smallest-system value. The dotted line in the all-input relative plot is an empirical linear fit to the measured data, not a theoretical equality; the dotted line in the fixed-input relative plot is the empirical mean, included to show whether the curve is approximately horizontal.

5.2 One-dimensional Heisenberg chain

The first experiment follows the numerical setting in the paper. The Hamiltonian is the normalized nearest-neighbor Heisenberg chain

$$H = \frac{1}{n-1} \sum_{i=1}^{n-1} (X_i X_{i+1} + Y_i Y_{i+1} + Z_i Z_{i+1}),$$

where $n = 4, 5, 6, 7, 8$ is the number of qubits. This normalization gives $\lambda = 3$ for all system sizes. The gate counts are $N = 20, 40, 80, 120, 160, 240, 320$, and the system-size comparison uses $N = 160$.

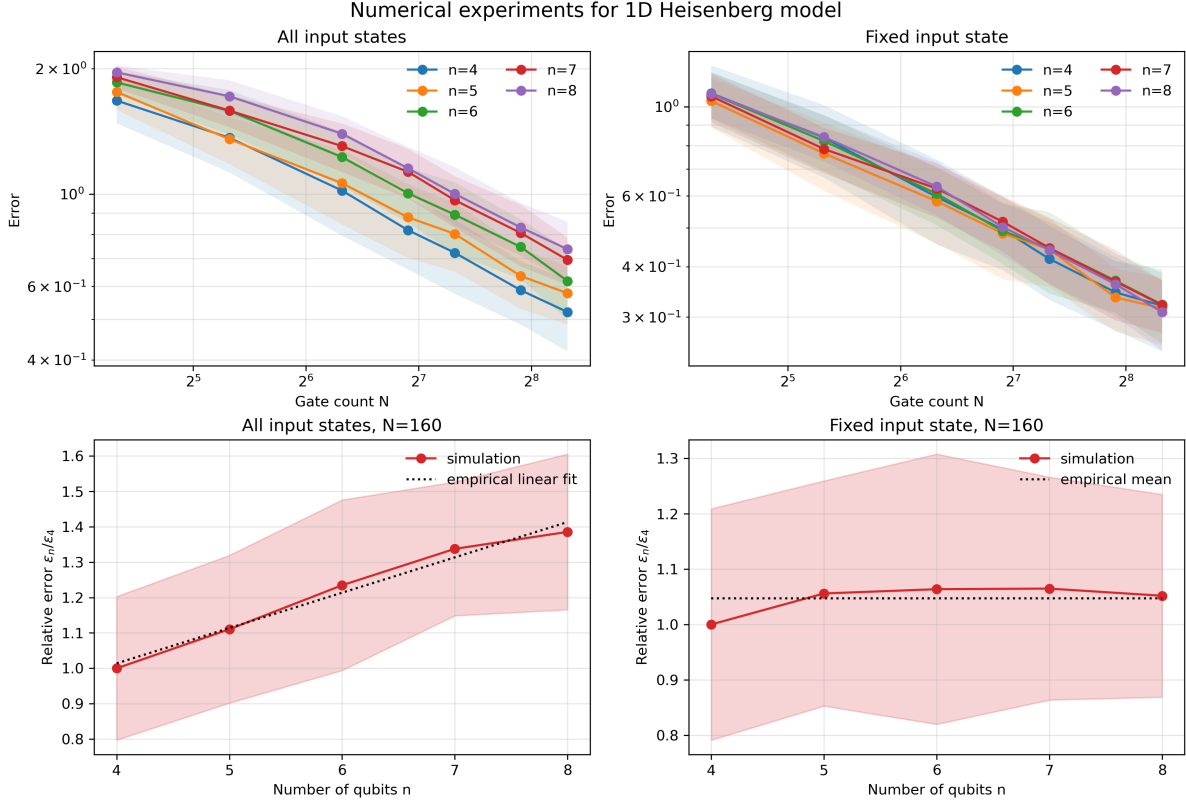


Figure 1: qDRIFT simulation of the normalized one-dimensional Heisenberg chain. The top row compares the decay of the error as the gate count N increases. The bottom row compares the normalized error as the number of qubits increases at fixed $N = 160$. The fixed-input error is much flatter than the all-input error, in agreement with the dimension-free fixed-input concentration bound.

The numerical behavior matches the theory qualitatively. The all-input error increases with the number of qubits at fixed N , reflecting the dimension factor in Theorem 1. By contrast, the fixed-input error remains close to a horizontal trend. This illustrates that the factor n is tied to controlling all possible inputs at once, rather than to the qDRIFT sampling procedure itself.

5.3 SYK model

The second experiment replaces the Heisenberg Hamiltonian by a Sachdev–Ye–Kitaev (SYK) Hamiltonian. For n Majorana fermions,

$$H = \sum_{1 \leq i < j < k < l \leq n} J_{ijkl} \chi_i \chi_j \chi_k \chi_l,$$

where the couplings are sampled from a centered Gaussian distribution with

$$\overline{J_{ijkl}^2} = \frac{3!J^2}{n^3}.$$

The Majorana operators are represented by the Jordan–Wigner construction

$$\chi_{2r+1} = \frac{1}{\sqrt{2}} Z_1 \cdots Z_r X_{r+1}, \quad \chi_{2r+2} = \frac{1}{\sqrt{2}} Z_1 \cdots Z_r Y_{r+1}.$$

Thus $n = 8, 10, 12, 14, 16$ Majorana modes correspond to 4, 5, 6, 7, 8 qubits. After drawing the Gaussian couplings, the Hamiltonian is rescaled so that $\lambda = 3$ for every system size. This extra normalization is important: without it, the qDRIFT strength λ grows with the number of SYK terms, and a fixed N comparison would mix system-size effects with a changing Hamiltonian strength.

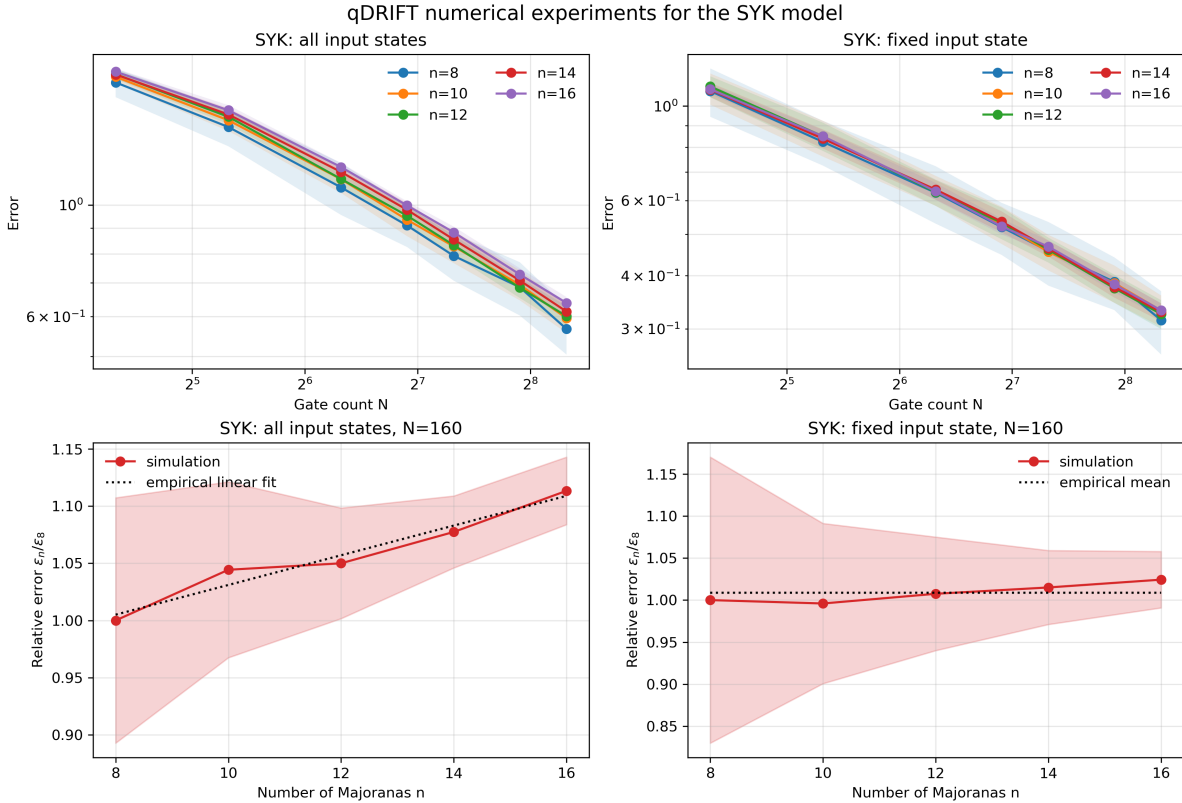


Figure 2: qDRIFT simulation of the SYK model after rescaling each instance to $\lambda = 3$. The same gate counts and error metrics as in the Heisenberg experiment are used. Once the Hamiltonian strength is fixed, the fixed-input error is again nearly independent of system size, while the all-input error shows a stronger size dependence.

The SYK experiment confirms that the fixed-input advantage is not special to geometrically local Hamiltonians. Even though the SYK Hamiltonian contains all four-Majorana interaction terms and is highly nonlocal after the Jordan–Wigner mapping, the fixed-input curve remains almost flat once λ is normalized. This is consistent with Theorem 2, whose

bound depends on λ , t , and N , but not explicitly on the Hilbert-space dimension. The all-input curve, on the other hand, remains more sensitive to system size, as expected from the matrix martingale bound.

6 Summary

Viewing QDRIFT as a random walk on $U(d)$, the error splits into a deterministic bias (controlled to $O(t^2\lambda^2/N)$ by matrix-exponential estimates and telescoping) and a random fluctuation (controlled to $O(\sqrt{n}t^2\lambda^2/N)$ by a matrix martingale). Restricting to a fixed input replaces the matrix martingale by a dimension-free vector martingale, removing n and giving an n -fold reduction with no knowledge of the state. The L -independence of QDRIFT thus stems from random sampling itself; averaging only upgrades accuracy from $1/\epsilon^2$ to $1/\epsilon$. Section 5 reproduces the qualitative scalings $E_{\text{worst}} \sim \sqrt{n/N}$ and $E_{\text{fixed}} \sim N^{-1/2}$ numerically on Heisenberg and SYK Hamiltonians.

References

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